🔑 **Answer Key — Teacher Copy · Lesson 6-14**

MSTAR PRACTICE WORKSHEET

Lesson 6-14: The Distributive Property

Grade 6 Mathematics · Standard 6.AT.7

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: A

Expand 6(n + 3) using the distributive property.

- **A** 6n + 18
- **B** 6n + 3
- **C** 6n + 9
- **D** n + 18

6(n + 3) = 6 × n + 6 × 3 = 6n + 18.

Question 2 SELECTED RESPONSE — CORRECT: B

Expand 4(5 – 2) using the distributive property.

- **A** 9
- **B** 12
- **C** 20 – 2
- **D** 22

4(5 – 2) = 4 × 5 – 4 × 2 = 20 – 8 = 12.

Question 3 SELECTED RESPONSE — CORRECT: A

Expand 3(x + 4) using the distributive property.

- (A) 3x + 12
- (B) 3x + 4
- (C) 7x
- (D) x + 12

3(x + 4) = 3 × x + 3 × 4 = 3x + 12.

Question 4 SELECTED RESPONSE — CORRECT: D

When using the distributive property with a(b + c), you must multiply 'a' by:

- (A) Only b
- (B) Only c
- (C) b + c first, then multiply
- (D) Both b and c

The distributive property says a(b + c) = ab + ac. You multiply the outside factor by EACH term inside the parentheses.

Question 5 SELECTED RESPONSE — CORRECT: A

Light Cue 1: Expand 3(x + 4) for the stage-light controller.

- (A) 3x + 12
- (B) 3x + 4
- (C) 3x + 7
- (D) x + 12

Distribute the 3 to each term: 3 × x + 3 × 4 = 3x + 12.

Question 6 SELECTED RESPONSE — CORRECT: B

Light Cue 2: Expand 5(2 + n) to set two spotlight banks.

A. 7 + 5n

B. 10 + 5n

C. 10 + n

D. 10 + 5

Distribute the 5 to each term: 5 × 2 + 5 × n = 10 + 5n.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: B

Which expression is equivalent to 6(2x + 5)?

A. 2x + 30

B. 12x + 30

C. 12x + 5

D. 12x + 11

The distributive property says a(b + c) = ab + ac. Multiply the outside factor by each term inside: 6 · 2x = 12x and 6 · 5 = 30, so the expression is 12x + 30.

- **A:** The 6 reached the 5 but not the 2x. Distribute it to both terms inside.
- **C:** The 6 reached 2x but not the 5. Distribute it to BOTH terms inside.
- **D:** The 6 and the 5 were added rather than multiplied. Six times five is 30.

Part B — correct answer: B

Which reasoning shows why your answer to Part A is correct?

A. Multiply the outside factor by the second term only and copy the first term: 2x + 30.

B. Multiply the outside factor by every term inside the parentheses: 6 · 2x = 12x and 6 · 5 = 30, giving 12x + 30.

C. Multiply 6 by 2x, then add 6 and 5 instead of multiplying them: 12x + 11.

D. Multiply the outside factor by the first term only and copy the second term: 12x + 5.

Expanding requires the outside factor to reach every term inside the parentheses. Skipping a term or adding instead of multiplying produces an expression that is not equivalent.

Question 8 WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Pulling Out a Smaller Factor

> Elena was asked to factor 18m + 24 completely. She wrote 18m + 24 = 2(9m + 12) and said she was finished because 2 divides both terms.

Explain Elena's misconception and give the completely factored expression.

Model answer: Elena pulled out a common factor instead of the greatest common factor. Her expression is equivalent to the original, but 9m and 12 still share the factor 3. The GCF of 18 and 24 is 6, so 18m + 24 = 6(3m + 4). Checking by expanding: 6 · 3m = 18m and 6 · 4 = 24.

2 points	Student explains that 2 is a common factor but not the greatest common factor of 18 and 24, so the expression is not completely factored, and gives 6(3m + 4).
1 point	Student gives 6(3m + 4) but does not explain that Elena used a common factor instead of the greatest common factor.
0 points	Student agrees that 2(9m + 12) is completely factored or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

Use the distributive property to explain why 6 × 98 can be calculated as 6 × 100 − 6 × 2 = 588. Then create your own example of using the distributive property to make multiplication easier.

Model answer: Because 98 = 100 − 2, you can rewrite 6 × 98 as 6(100 − 2) and distribute: 6 × 100 − 6 × 2 = 600 − 12 = 588. The distributive property is what makes that rewrite legal, and it turns one hard multiplication into two easy ones — the same trick works for problems like 7 × 99 or 4 × 52.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.